

Homework 10

Due April 19th

For both problems on this homework, assume the same amplifier and that the circuits are at 25 °C:

- **for -4050:** Assume the amplifier has $\text{GBP} = 3 \text{ MHz}$, $e_{nw} = 20 \text{ nV}/\sqrt{\text{Hz}}$, $f_{ce} = 1 \text{ kHz}$, and $i_{nw} = 0$
- **for -6050:** Assume the amplifier is a [AD8613](#). Extract the necessary parameters for this amplifier from the datasheet necessary for characterizing AC operation and noise. All parameters must be taken either from the specifications table (use Table 1, $V_{SY} = 5 \text{ V}$, $R_L = 100 \text{ k}\Omega$) or estimated from the graphs. For the case of voltage noise, consider Figure 25. When estimating the voltage noise, assume that it follows the characteristic equation as given in the lecture. Since there is no current noise graph provided, you may assume that it only white noise: $f_{ci} = 0$.
- Other notes: assume that thermal noise dominates the noise produced by any resistances. To integrate the frequency range to compute E_{no} , you may either (1) do it analytically, (2) numerically integrate using software (e.g. Matlab), or (3) estimate the integration by hand assuming poles/zeros are ideal (straight line graphs). For whichever method selected, ensure to document the process.

Problems:

1. **-6050 Only:** Perform the parameter extraction from the datasheet as required above.
2. A summing amplifier is constructed to produce the output $v_o = -(v_1 + v_2/2 + v_3/3)$, where the feedback resistor for the circuit is $10 \text{ k}\Omega$. What is the total noise on the output, E_{no} , for the circuit between 10 Hz and 100 MHz?
3. Analyze the noise characteristics of the higher sensitivity I-V converter analyzed in lecture 5 (given below). Specifically, find the total noise on the output for all frequencies above 0.1 Hz. Using this total noise value, evaluate the SNR for the circuit for if the input, I_{in} has a $10 \mu\text{A}$ amplitude and is a (a) sine wave, (b) square wave, and (c) triangular wave. Use $R_1 = 20 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $R_3 = 15 \text{ k}\Omega$.

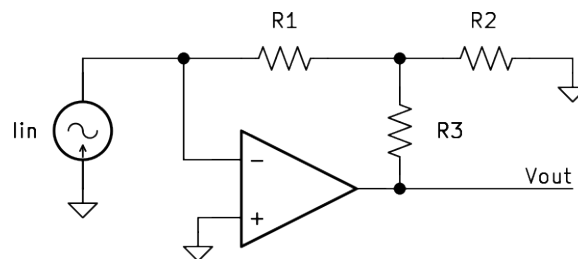


Figure 1: Question 2 circuit